Response to Reviewer 1

Manuscript ID 94388: FedRBC: Blockchain-Enabled Reputation-Weighted Robust Federated Learning for Privacy-Preserving Healthcare IoT Edge Networks

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Dear Reviewer 1,

We thank you for a careful and constructive review — your eighteen points have materially improved the paper. In response, we extended the threat model with four new attack families (LIE, backdoor with attack-success-rate scoring, root-set poisoning, and direction-preserving scaling under differential privacy), raised every headline configuration to ten seeds with 95% confidence intervals and Holm-corrected significance tests, added the classical baselines and the upper references you asked for, restructured the convergence theory so that its residual adversarial term is derived explicitly from the gate and ledger rather than assumed, measured the full component-ablation grid including the ledger-memory ablation, expanded the sensitivity and client-scaling studies, and widened the deployment, false-quarantine, and clinical-safety discussion. We respond point by point below; all section/equation/figure/table numbers refer to the revised manuscript.

R1.1 — Novelty should be more clearly distinguished from existing trust-based aggregation, reference-guided robust learning, and blockchain reputation mechanisms

Response. We agree the positioning needed sharpening. The revised Section 1.2 now states the separation along three axes: (i) against root-referenced robust learning (FLTrust), FedRBC is not a different gate function but a *composition* — a temperature-controlled softmax gate that keeps skewed-but-benign non-IID updates (which ReLU truncation discards; Section 5.7 quantifies this tax), plus Layer-0 magnitude pre-filtering, relative clipping, and persistent on-chain evidence; (ii) against trust-based aggregation, reputation in FedRBC is not a score computed and forgotten per round — it is committed on-chain, decays when idle, drives both aggregation weights and client selection, and actuates network quarantine; (iii) against blockchain trust systems, the ledger stores only 5 bytes of scalar evidence per client-round (no gradients or identifiers), keeping the chain inside the privacy envelope of the FL protocol itself. The new Lemma 1 (Eq. 15) makes the composition claim precise: the residual malicious weight is bounded by the *product* of a gate factor $e^{-\Delta c/T_s}$ and a ledger factor ρ_r that contracts geometrically (Prop. 1) — neither factor is available to any single-mechanism design.

Changes. Section 1.2 (three-axis positioning); Section 4.6 (Lemma 1); Contribution C1.

R1.2 — Stronger justification for root dataset construction, representativeness, updating frequency, and resistance to poisoning; experiments on a compromised or outdated reference dataset

Response. (1) *Construction and representativeness*: the root set $\mathcal{D}_r$ is a server-held stratified sample of 500 flows drawn from the same preprocessing pipeline as the training partitions (Section 4.1); its role is directional anchoring, not training — it enters Eqs. 4–5 only through the direction and norm of Δ_{root}. We evaluate sensitivity to root size in the new sweep (root sizes 100 and 300 against the default 500). (2) *Updating*: the root update is recomputed every round; we discuss refresh cadence against distribution drift in Section 6. (3)

Resistance to poisoning: the root set never receives client updates and is not reachable by training-plane clients — the attack surface is server/ledger compromise, which is precisely the scenario of the new root-perturbation study. We now report F1 as the root direction is adversarially perturbed at levels $\varepsilon \in 0.1, 0.2, 0.3$ for FedRBC and FLTrust under both Sybil and sign-flip attacks: results in the new Section 5.9. (4) The theory now states the premise explicitly: Lemma 1's directional-separation assumption presumes an untampered root set, and the revised Section 4.6 states that the poisoned-root experiments measure exactly the degradation when this premise fails — the security envelope of the bound is now explicit rather than implicit.

Changes. Section 4.6 (Scope paragraph); new Section 5.9 (root-perturbation study); Section 6 (refresh cadence discussion); sensitivity sweep on root size.

R1.3 — Evaluate adaptive model-replacement attacks, backdoor attacks, stealthy poisoning, and direction-preserving magnitude manipulation, or delimit the scope of security claims

Response. We did both. (i) *Scope*: Section 4.6 now contains an explicit "Scope of the guarantees" paragraph — the theory applies constructively to direction-inconsistent attacks, and direction-preserving adversaries shrink the separation gap Δc toward zero, degrading the exponential gate factor; the defence then rests on Layer-0 filtering, relative clipping, and accumulated reputation evidence. (ii) *New experiments* (Section 5.9): the LIE attack of Baruch et al. (collusive per-coordinate mean-plus-z-sigma perturbation, which stays direction-aligned and evades norm filters); a targeted backdoor with trigger injection scored by attack success rate (ASR) alongside main-task F1; the direction-preserving scaling attack already in Section 5.7; and root-set perturbation per R1.2. Together these cover the four families requested.

Changes. Section 4.6 (scope); new Section 5.9 (LIE, backdoor+ASR, root poisoning); Section 5.7 (scaling).

R1.4 — Verify that the convergence bound follows from the proposed aggregation rule under the specified Byzantine threat model, including highly heterogeneous non-IID data

Response. The theorem has been restructured precisely to meet this requirement. In the previous version the residual malicious weight $\bar{\epsilon}_r$ was an assumption — as Reviewer 2 also noted, this is the very quantity the gate should bound. The revised Section 4.6 removes the circularity in two steps: **Lemma 1** derives $\epsilon_r(t) = \sum_{i \in \mathcal{B}} q_i(t) \leq \rho_\beta \rho_p \rho_r e^{-\Delta c / T_s}$ (Eq. 15) directly from the softmax gate (Eq. 6) and the ledger bounds (Eqs. 8–9) under three *measurable* conditions (directional separation $\Delta c > 0$, honest-preservation of the Layer-0 filter, bounded priors and reputations); **Theorem 1** then consumes this explicit quantity in a standard descent argument. Every constant is derived in Appendix B.1. The bound's heterogeneity handling is the standard bounded-variance assumption σ^2; the heterogeneity *sensitivity* is measured empirically by the Dirichlet sweep down to $\alpha = 0.1$ (Section 5.4), where FedRBC degrades gracefully while plurality-based baselines collapse. The premises themselves (Δc, ρ_r, ϵ_r) are logged per round in the released code, so the theorem's conditions are auditable rather than taken on faith.

Changes. Section 4.6 (Lemma 1 + Theorem 1 restructured); Appendix B.1 (full proof of Lemma 1 and Theorem 1); Section 5.4 (non-IID sweep).

R1.5 — Systematic sensitivity analysis under different parameter configurations and operational conditions

Response. The revision adds a systematic sweep over every mechanism parameter: gate temperature T_s, relative-clip scaling κ_c, evidence threshold θ_h, quorum size K, and root-set size (100/300 against the default 500); plus the operational conditions of Byzantine ratio $\beta \in [0.1, 0.5]$ (Section 5.4 and the new β=0.5 extension of R2.3) and Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$ (Section 5.4). All sweep points are 3-seed means with std reported. The sweep (Section 5.9) locates the defaults near local optima and its profile is itself informative: the gate temperature is deliberately flat across $T_s \in \{0.1, 0.5\}$ (confirming that the softmax form distinguishes by *degree* rather than a knife-edge threshold), the reward threshold and quorum size are mildly monotone, and the single genuinely harmful setting is over-tight clipping ($\kappa_c = 1$, about 6 points) — the one parameter the asymmetry analysis of Section 6.2 warns about.

Changes. Sensitivity sweep in Section 5.9 (T_s, κ_c, θ_h, K, root size; each point 3 seeds with std); Section 5.4 (β sweep, with the β=0.5 extension in Section 5.9).

R1.6 — More independent runs, confidence intervals, and statistical analyses across client populations and network conditions

Response. All headline configurations are now **10 seeds** (up from 3–5); every mean carries ± std and a 95% confidence interval; the nine pairwise comparisons against baselines now carry **Holm's step-down correction** for multiple testing, and the corrected p-values are quoted in the text alongside Table 4. Per-seed values are disclosed in the reproducibility package, including the collapsed seeds (see R2.5 below — we report and explain them, we do not hide them). Client populations: the new client-scaling study evaluates N = 20, 50, 100 with the same per-population statistical treatment.

Changes. Main results table (10 seeds, 95% CI, Holm-corrected p); per-seed disclosure; client-scaling study (Section 5.9).

R1.7 & R1.8 — Baseline reimplementations (RAID, FedGT) may affect fairness; validate implementations and separate full-system from simplified-variant comparisons

Response. We accept this criticism and have handled it in three ways. (i) *Disclosure*: Section 5.2 now labels RAID and FedGT entries explicitly as aggregation-level re-implementations against FedRBC's own root reference (an internal ablation of the design space), not as full-system reproductions; the text no longer claims system-level superiority over them on their native ground. (ii) *Faithful classical baselines added*: FLTrust, Krum, and trimmed mean — canonical mechanisms with unambiguous definitions — are now implemented, tuned, and reported in the main comparison, and FLTrust is additionally run at 10 seeds for a like-for-like comparison with FedRBC. (iii) *Upper bounds*: centralized training and clean FedAvg now anchor the comparison (see R2.1). The comparison structure the reviewers asked for — full-system FedRBC vs. canonical mechanisms vs. clearly-labeled ablated variants vs. upper bounds — is exactly what Tables 3–4 now present.

Changes. Section 5.2 (baseline taxonomy and disclosure); Tables 3–4 (FLTrust/Krum/trimmed-mean/upper bounds); FLTrust 10-seed comparison.

R1.9 — Investigate seed instability; provide detailed per-seed results; explain effects of downsampling, class imbalance, and partitioning

Response. The instability had a specific, now-fixed root cause, and we document it transparently. In the submitted version, a single no-attack seed collapsed to F1 = 0.000. Tracing the logs showed a cosine-similarity 0/0 case producing a NaN that propagated through the reputation vector into the model update (this is also why several std values were exactly 0.000 — degenerate single-class predictions of collapsed seeds). The fix is defensive and now part of the released code: (a) the update application rejects any non-finite aggregate; (b) evidence and reputation scores are NaN-guarded. After the fix the affected seed recovers to F1 = 0.454, and **all** reported numbers in the revision come from the fixed pipeline (data tag 'rev1'/'v5'). Per-seed tables are in the package. We additionally report the dual-collapsed-seed pattern of the simplified FedGT/RAID variants and explain why their accuracy stays at the majority-class rate while F1 collapses (Section 5.3). The dataset effects are analyzed in Section 5.1: the 12.5% attack class bounds achievable recall (the centralized upper bound of 0.665 confirms the 0.5–0.6 band is the task's separability ceiling, not a defence failure), and the balanced downsampling of CICIoT2023 (3:1) with its rationale is documented for the second dataset.

Changes. Section 5.7 (root-cause and numerical-stability paragraph); Section 5.3 (separability-ceiling calibration); Section 5.8 (CICIoT downsampling rationale); Section 5.3 (collapsed-seed analysis); per-seed tables in package; all numbers re-run post-fix.

R1.10 — Discuss gradient-inversion and membership-inference attacks; evaluate differential privacy or secure aggregation and their effects on detection accuracy and robustness

Response. The revision adds a Gaussian output-perturbation layer on the aggregated step (per-client updates are already clipped, so the aggregate's sensitivity is bounded; per-round noise at $\sigma_{\text{dp}} \in \{0.005, 0.02, 0.1\}$ scaled by the reference norm, with accountancy via RDP composition stated as an (ε, δ) budget per training job — new Section 4.7, Eqs. 16–17), and measures its effect on *both* detection accuracy and Byzantine robustness for FedRBC, FLTrust, EIFFeL, and FedAvg under sign-flip and Sybil attacks. The composition question — does privacy noise blunt the directional gate? — is answered empirically (Table 7): at $\sigma_{\text{dp}} = 0.005$–$0.02$ the perturbation costs FedRBC at most a few points (0.608/0.549 and 0.585/0.510 F1 under sign-flip/Sybil, essentially its no-DP means), and at the stress level $\sigma_{\text{dp}} = 0.1$ — noise magnitude rivaling the reference norm — FedRBC retains 0.447–0.448 while FLTrust collapses to 0.220–0.222: a hard ReLU gate fed noisy evidence zeroes out exactly the honest mass it should keep, while the softmax gate degrades gracefully. Because the noise is injected after evidence scoring (Section 4.7), the gate's inputs are clean by construction. Gradient-inversion and membership-inference threats are discussed in Section 6 with citations, including why the root set is never trained on and stays server-held.

Changes. New Section 5.9 (DP study over 4 methods × 2 attacks × 3 noise levels); Section 4 (DP mechanism description with RDP budget); Section 6 (inversion/membership-inference discussion). The title's "Privacy-Preserving" is now backed by an accounted mechanism (see also Reviewer 2, "Also required").

R1.11 — Evaluate transaction delays, throughput, gas consumption, controller failures, communication overhead, and consequences of compromised infrastructure

Response. Section 5.6 already reports per-round wall-clock overhead, gas per evidence write, and end-to-end containment latency; the revision extends it with (i) throughput saturation analysis — with 10 writes/round at 48.6k gas each, the ledger write path saturates near 150 participating clients per round, which we state as an

explicit scaling boundary and connect to the client-scaling study; (ii) a controller-failure discussion in Section 6 (ONOS unavailability degrades containment but never the learning path — the model keeps training under FedRBC's aggregation-side defences, which is an architectural property); (iii) communication overhead per round — model-update payloads under TLS plus the 4-byte evidence and 1-byte reputation scalars on-chain (about 25 KB of ledger data per training job) — documented in the revised Section 5.6 text and the payload classes of Section 4.1.

Changes. Section 5.6 (gas-saturation boundary, emulation labelling, communication overhead); Section 6 (controller failure, compromised-infrastructure discussion).

R1.12 — Justify MTTC assumptions; compare theoretical estimates with empirical measurements; distinguish theoretical vs observed containment latency

Response. Three specific changes. (i) The Fig. 9 caption and Section 5.6 now state explicitly that the cloud-queueing curve is an M/M/1 emulation whose parameters (arrival/service rates) are set from the emulated workload, not a measurement of a production cloud — the abstract's 50–80% claim is correspondingly qualified as a separation achieved under the emulated parameterization. (ii) The provenance is now labelled per component: the edge path's flatness (about 3.3 s — deterministic block commit plus local rule installation) is a measurement of the emulated testbed, while the cloud path's divergence is an M/M/1 emulation with parameters set from the emulated workload, so the theory–measurement correspondence is explicit rather than implicit. (iii) Section 5.6 notes that the separation is robust across utilization because the edge path has no queue to grow, and points to the GI/G/1 limitation for bursty, heavy-tailed arrivals in Section 6.3, so the claim's parameter dependence is visible rather than implicit.

Changes. Fig. 9 caption; Section 5.6 (theory vs emulation labelling + sensitivity note); Abstract (qualified MTTC claim).

R1.13 — Independently remove directional gating, magnitude clipping, reputation weighting, persistent blockchain storage, selection, and SDN containment to quantify each component's contribution

Response. The revision contains the full component-ablation grid, each component removed independently under all three attack classes plus no-attack (3 matched seeds each, same-seed full-system reference; Table 8): gate, clip, Layer-0, reputation weighting, selection, and the new cross-round-memory switch (per-round reputation reset — the ledger-memory ablation Reviewer 2 asked for in C2). The pattern is attack-dependent and honest: selection contributes most under sign-flip and label-flip because its absence occasionally admits a malicious plurality into the quorum (random selection collapses: 0.497 ± 0.212 and 0.503 ± 0.226); relative clipping is decisive under sign-flip (-8.2 points, the one attack it exists to bound); gate, Layer-0, and reputation weighting contribute 1–5 points each with attack-dependent sign; and the no-memory ablation leaves final F1 statistically unchanged (0.564/0.604/0.610 vs 0.602/0.581/0.578 same-seed full) — which *sharpens* the ledger's case rather than weakening it: the F1 defence is carried by the gate, while the ledger's contribution is exactly the operational properties an aggregation rule cannot express (the $12\times$ round-60 reputation separation of Section 5.5, resistance to consequence-resetting intermittent adversaries, the immutable audit trail, and containment). SDN containment is evaluated in the MTTC path (Sections 4.5/5.6) — its removal converts quarantined clients into merely down-weighted ones, which is quantified by the selection ablation's degradation under persistent re-attack; we state this linkage explicitly.

Changes. New ablation grid (Table 8); no-memory ledger ablation; Sections 5.7/5.9.

R1.14 — Ensure equations, algorithms, parameters, and implementation are fully consistent

Response. A symbol-level audit was performed across the manuscript: the clipping coefficient is now κ_c everywhere (Eq. 5, Alg. 1, Table 2, Theorem 1, Appendix B.1), deliberately distinct from the penalty-acceleration product $\kappa\gamma$ of Eq. 9 — the β/κ collision Reviewer 2 identified is fixed at its root; equation numbering is now strictly ascending (1)–(15) with all references verified programmatically; Table 2's parameter values match both the algorithms and the released configuration files; and the residual-malicious-weight symbol ϵ_r is defined once (Lemma 1) and consumed consistently in Theorem 1 and Appendix B.1.

Changes. Global symbol audit (κ_c rename in 8 places); Eq. numbering 1–15; Lemma 1/Theorem 1 consistency.

R1.15 — Discuss device compatibility, resource-constrained deployment, false quarantines, recovery, and clinical consequences of isolating equipment

Response. Section 6 (Discussion) now covers: (i) compute envelope — FedRBC's client-side cost is local SGD plus a cosine with the broadcast root direction, no crypto operations on the device; (ii) *false-quarantine* — measured and reported honestly, not claimed away: in the no-attack runs, 75–100% of clients touch the quarantine line at least once over 60 rounds, because directional evidence becomes noise-dominated as update norms shrink late in training, so grey-zone penalization erodes honest reputations as well as malicious ones (Section 5.5). We explain why this does not break the system — the relative ranking survives (round-60 honest median 0.109 vs Byzantine 0.009 across the attack runs), the F1 defence is carried by the gate (the no-memory ablation of Section 5.9 shows resetting all reputations leaves final F1 unchanged), and cold-start probation is the recovery path — but we now state plainly that r_{q} must be read as a *relative-discrimination* threshold rather than an absolute no-false-quarantine guarantee, and we name the direct fix (a penalty gate that widens the grey zone as $\|\Delta_{\mathrm{root}}\|$ shrinks) as future work; (iii) *recovery* — quarantined devices re-enter through cold-start probation at r_0 with a documented path (Section 4.4/4.5); (iv) *clinical safety* — containment isolates the device's *training traffic* from the federation, not its clinical function; we state this scope explicitly and discuss read-only fallback modes for critical equipment.

Changes. Section 6 (deployment, false-quarantine, recovery, clinical-scope paragraphs).

R1.16 — Provide source code or a reproducibility package; document deviations from original baseline implementations

Response. The complete package is openly released; the repository URL is given in the Availability of Data and Materials statement at the end of the paper, and Section 5.1 now carries an explicit in-text pointer to it. The package contains all training/attack/aggregation code, per-seed raw results (JSON + per-round CSV), figure-generation scripts that read only the released data, the LaTeX source, and a README that reconstructs every table and figure from raw files with one command per artifact. Deviations of the RAID/FedGT re-implementations are documented in code comments and in Section 5.2's disclosure paragraph.

Changes. Availability of Data and Materials statement (repository URL and package contents); Section 5.1 (in-text availability pointer); Section 5.2 (deviation notes).

R1.17 — Moderate claims of deployment readiness and broad attack resistance

Response. Done throughout: "deployable" language is confined to the emulated testbed (Reviewer 2's related point about 20 logical clients is addressed the same way — the evaluation is now explicitly labelled an emulation throughout); the abstract's robustness claim is scoped to the evaluated attack classes with the new families named; the theory's scope paragraph (Section 4.6) states which attacks the gate bound covers and which fall to the statistical layers; and the MTTC claim is qualified per R1.12.

Changes. Abstract; Section 4.6; Section 5 (emulation labelling); Section 6.

R1.18 — Verify notation, improve figure readability, ensure tables/figures correspond to experiments, check references

Response. Notation audit per R1.14. For figure readability, Figs. 1–3 were re-rendered for this revision with enlarged label fonts and numbered-flow badges (the swim-lane style of the submitted version is retained), and figure titles are carried by the captions rather than in-figure text; the legends of Figs. 6–7 were repositioned into whitespace (Editor's E4) and the shaded regions explained in captions (E5: the band of Fig. 7 is the mean±std band over the 5 seeds; the P10–P90 band appears in Fig. 8's reputation trajectories); the reproducibility package documents the exact data tag each figure is generated from, and figure-number references were re-verified against the experiments; references re-verified against CrossRef (two title corrections, one completed venue entry — Editor's E8).

Changes. Figs. 1–3 (re-rendered, larger labels/badges); Figs. 6–7 (legends), Fig. 7 (band statistic in caption); references (E8); global notation pass.